

Entropy of a full-support CCD

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We estimate the phylogenetic entropy of the full-support CCD regulariser of the companion note. The standard CCD entropy is the wrong quantity here; we give two estimators that agree with brute force where it is computable, apply them to the RSV2 posterior (129 taxa), and use them to track how the escape parameter μ should change with sample size.

Why the standard CCD entropy fails

For a plain CCD every tree probability is a product of conditional clade probabilities (CCPs) θ over a finite set of observed clades, and the entropy follows the Lewis recursion $H = -\sum_C P(C) \sum_{(L,R)} \theta(L,R | C) \log \theta(L,R | C)$. The full-support model breaks this assumption: at a reservable clade the conditional distribution is a *mixture* of an observed (red) split with probability $1 - \mu$ and an escape into a *blue region* (novel clades) with probability μ , and the observed CCPs sum to $1 - \mu$, not 1. The Lewis sum ignores the $(1 - \mu)$ discount, the escape mass, and the entire novel-clade support, so it returns the entropy of the renormalised observed backbone — a systematic underestimate. We therefore do not use the standard formula, and estimate the entropy directly.

Two estimators

We write $P(T)$ for the model probability of a tree: each red node contributes $(1 - \mu) \theta(S_v | C_v)$ and each maximal blue region with top clade C and $m - 2$ novel clades contributes $\epsilon(C)^{m-2}$ (FLAT weighting; SHARED divides by the boundary's pathcount). Here $\epsilon(C)$ solves $\sum_{j \geq 1} N_j(C) \epsilon^j = \mu$ over the computed reserve orders $j \leq k$, with N_j the number of order- j boundaries.

Sampling estimator. Entropy is the expected surprisal, so

$$\hat{H} = -\frac{1}{S} \sum_{s=1}^S \log P(T_s), \quad T_s \sim \text{model}, \quad \text{SE} = \text{sd}(\log P) / \sqrt{S}. \quad (1)$$

The trees T_s are drawn from the model itself and each carries its probability $P(T_s)$, so the estimate is unbiased; its only cost is the $O(1/\sqrt{S})$ sampling noise.

Recursive estimator (generalised Lewis recursion). The model is generated by a recursive branching process over the observed-clade DAG, so entropy decomposes by the chain rule. Let $H_{\text{free}}(C)$ be the entropy of the subtree generated when C may escape, and $H_{\text{red}}(C)$ when C is forced to take an observed split (a

blue region’s boundary part). Then

$$H_{\text{red}}(C) = \sum_{(L,R)} \theta(L, R \mid C) \left[-\log \theta(L, R \mid C) + H_{\text{free}}(L) + H_{\text{free}}(R) \right], \quad (2)$$

$$\begin{aligned} H_{\text{free}}(C) = & \underbrace{(1 - \mu) \left[-\log(1 - \mu) + H_{\text{red}}(C) \right]}_{\text{red branch}} - \underbrace{\log \epsilon(C) \sum_m (m - 2) N_m \epsilon(C)^{m-2}}_{\text{blue self-information}} \\ & + \underbrace{\sum_{\text{regions}} P(\text{region}) \sum_i H_{\text{red}}(B_i)}_{\text{blue children}}, \end{aligned} \quad (3)$$

with $H_{\text{free}}(C) = H_{\text{red}}(C)$ when C reserves nothing. The red part and the blue self-information are closed-form; the blue-children term is summed exactly by enumerating the same canonical boundaries the reserve uses (FLAT weights each by its pathcount). Computing $H_{\text{free}}, H_{\text{red}}$ for clades in increasing size makes every child and boundary part already available, so the recursion is deterministic and *exact* for the self-consistent distribution (escape mass μ , regions up to depth k).

Which distribution. The entropy reported is that of whichever distribution one chooses to draw from. The *self-consistent* distribution (escape mass exactly μ , blue regions only up to reserve depth k) is normalised and has the closed-form recursion above. The *full-support* distribution additionally carries the reserve tail (regions deeper than k); this tail is the same intractable quantity that can only be estimated by sampling, so the full-support entropy is obtained from the sampling estimator rather than the recursion. The two agree only when the chosen distribution and the tail correction are matched: no tail with the self-consistent distribution, the sampled tail with the full-support distribution.

Validation against brute force

On four taxa with no tail correction the model normalises exactly, so summing $P(T)$ over all rooted topologies gives the exact entropy. The recursion matches it to 10^{-9} and the sampling estimate to within its standard error, for both weightings.

mode	μ	exact H	recursive	sampling ($\pm$ SE)
FLAT	0.05	1.033235	1.033235	1.0307 ± 0.0017
SHARED	0.05	1.023676	1.023676	1.0269 ± 0.0017
FLAT	0.01	0.784942	0.784942	0.7835 ± 0.0011

RSV2 (129 taxa)

The topology space is far too large to enumerate, so the two estimators cross-validate each other. At $\mu = 0.01$ (FLAT, $k = 2$) they agree, and the escape/novel-clade support adds about 11% over the observed-only backbone that the standard formula would report.

trees	clades	backbone (obs. only)	recursive	sampling ($\pm$ SE)
1,000	1143	51.23	59.19	59.02 ± 0.11
5,000	1630	52.79	59.31	59.17 ± 0.08
18,001	2009	53.31	59.10	59.08 ± 0.08

On the full posterior the estimators agree to 0.24 standard errors. (Entropies in nats.)

Selecting μ : dependence on sample size

The entropy grows with μ , so a fixed default is arbitrary: μ should be the value that maximises cross-validated held-out tree log-probability. On RSV2 this optimum is essentially independent of the backbone smoothing α (the same grid point wins for every α), so we fix α and grid μ . Taking disjoint, consecutive post-burn-in blocks of growing size, μ^* falls steadily: a richer backbone observes more clades, held-out trees contain fewer genuinely novel clades, and less escape mass is justified.

block	clades	μ^*	held-out (nat/tree)	$P(\text{novel})$	$H(\mu^*)$
128	730	0.02020	−62.42	0.80	60.78
512	926	0.00553	−56.40	0.34	56.04
2048	1278	0.00196	−55.37	0.13	55.57
8192	1743	0.00061	−54.40	0.041	54.78

Over the $64\times$ span μ^* drops by $\sim 33\times$, i.e. $\mu^* \propto N^{-0.8}$, a little slower than $1/N$; $P(\text{novel})$ collapses in step. All four optima are interior. At these small μ^* the reserve tail ($\sim \mu^3$) is negligible, so the self-consistent recursion and the full-support entropy coincide.

Held-out probability is a third entropy estimate

The held-out average log-probability is, by construction, the negative *cross-entropy* between the true posterior Q and the fitted model P :

$$-\overline{\log P}_{\text{held-out}} = H(Q, P) = H(Q) + \text{KL}(Q\|P). \quad (4)$$

The reported $H(\mu^*)$ is the model’s *self-entropy* $H(P)$; the two differ only in whether surprisal is averaged under Q or under P , so they coincide when $P \approx Q$ — which is exactly the regime μ^* targets, since maximising held-out log-probability minimises $\text{KL}(Q\|P)$ ($H(Q)$ being independent of μ). This is why the last two columns above nearly match, and it gives three estimates of the same posterior entropy (≈ 55 nats): the model self-entropy, the held-out cross-entropy (an upper bound), and brute force on small trees.

The residual gaps $+1.64, +0.36, -0.20, -0.38$ are interpretable. At small N the model fitted on a training subset carries a real $\text{KL} > 0$, lifting the cross-entropy above $H(P)$. As N grows $\text{KL} \rightarrow 0$ and a small *negative* gap remains: the regularised model is genuinely flatter (higher entropy) than the posterior. This is not a comparison artifact — measuring the held-out cross-entropy against the *same* cross-validation models’ self-entropy (rather than the whole-block model’s) gives essentially the same gap, ≈ -0.44 nats at $N = 8192$. Two effects make the model flatter, in very unequal measure: (i) it over-reserves novel-clade mass — model $P(\text{novel}) = 0.041$ against a held-out rate of 0.034, a 20% excess — but the binary entropy of that mismatch is only ≈ 0.02 nats; (ii) conditional on a novel clade, the model spreads its mass nearly uniformly over an astronomically large set of resolutions (high conditional entropy), whereas the posterior concentrates on the few novel clades actually visited, which accounts for the remaining ≈ 0.43 nats. The flatness is therefore mostly honest agnosticism about *which* unseen tree, not miscalibration of the *total* novel mass — the intended behaviour of a full-support prior.

Weighting and sample size. The figures above use the FLAT (clustering-neutral) weighting. The SHARED weighting shares a boundary’s escape mass over its pathcount resolutions, penalising *clustered* novel clades. On the 8192-tree block SHARED is indeed the better fit, but only marginally: held-out log-probability higher by 0.0033 nat/tree, and entropy at μ^* essentially unchanged (54.783 vs FLAT’s 54.785). The reason is that at large N the held-out-optimal $\mu^* \approx 6 \times 10^{-4}$ is small enough that escape is almost entirely *single* novel clades — the order at which FLAT and SHARED coincide — so the two weightings, which differ only on clustered novelty (mass $\sim \epsilon^2$), are near-identical here. The distinction bites at smaller N (larger μ^*); at the large- N operating points the entropy figures reported here are weighting-independent to three decimals.